

Group Members:

Muneha Adan (21L-6112)

Sania Nisar(21L-6065)

a) Kaggle account ID:

1) sanianisar1060

2) munehaadan

b) Rank on the leaderboard:**Leaderboard**[Raw Data](#)[Refresh](#)

This leaderboard is calculated with all of the test data.

#	Team	Members	Score	Entries	Last
1092	SaniaMuneha Best Team		0.97014	5	17h
1101	Muneha Adan		0.96975	5	37m

c) Methods used and results obtained:

Data Preprocessing: Normalized pixel values to [0, 1] and split data into 80% training and 20% validation sets.

Model 1: Single-layer neural network

- **Architecture:** Input layer (784 features) Output layer (10 neurons, softmax)
- **Optimizer:** Adam
- **Epochs:** 10

Model 2: Two hidden-layer neural network with dropout

- **Architecture:** Input layer - 256 neurons (ReLU) - 30% dropout - 128 neurons (ReLU) - 30% dropout - Output layer (10 neurons, softmax)
- **Optimizer:** Stochastic Gradient Descent (SGD)
- **Epochs:** 15

Results Obtained

- **Model 1:**
 - **Validation Accuracy:** 0.92
 - **Observation:** Underfitting, insufficient complexity
- **Model 2:**
 - **Validation Accuracy:** 0.97
 - **Observation:** Improved performance, better generalization with dropout

d) Error Analysis

- **Model 1:** Struggled with visually similar digits, and poorly written characters.
- **Model 2:** Reduced errors but still faced challenges with handwriting issues.